

2. Database Design & Data Modeling

This section presents the database design of the Restaurant Finder system. The design focuses on managing users, their favorite restaurants, and their custom reviews.

The system integrates with the Yelp API as an external data source. As a result, restaurant data is not stored locally in the database. Instead, restaurants are referenced using unique Yelp identifiers.

This approach helps reduce data redundancy and ensures that restaurant information such as ratings, menus, and opening hours is always up to date.

2.1 Entity-Relationship Diagram (ERD)

The Entity-Relationship Diagram (ERD) illustrates the core entities of the system and the relationships between them.

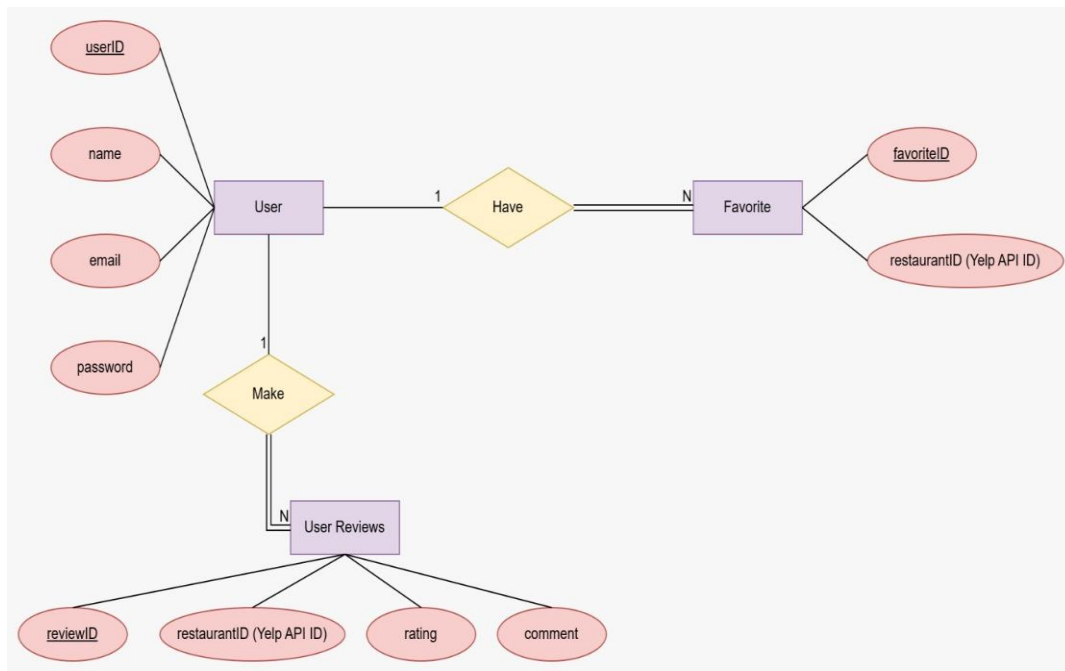
The system consists of the following main entities:

- **User:**
Represents the system user and contains basic information such as userID, name, email, and password.
- **Favorite:**
Represents the restaurants that a user marks as favorite. Each record is linked to a specific user and references a restaurant using the Yelp API ID.
- **Custom Review:**
Represents reviews written by users for restaurants. Each review includes a rating and a comment, and is associated with both a user and a restaurant.

Relationships

- A **User** can have multiple **Favorite** records (*One-to-Many*).
- A **User** can write multiple **Custom Reviews** (*One-to-Many*).

Each favorite and review does not store full restaurant details. Instead, it references restaurants using the Yelp API ID, allowing the system to fetch restaurant data dynamically from the external API.



2.2 Logical & Physical Schema

The logical and physical schema translates the ER Diagram into structured database tables. The system is composed of three main tables:

- **USER Table:**
Stores user information such as:
 - userID (Primary Key)
 - name
 - email
 - password
- **FAVORITE Table:**
Stores user-selected restaurants and includes:
 - userID (Foreign Key referencing USER)
 - yelp_rest_id (Restaurant identifier from Yelp API)
- **CUSTOM_REVIEW Table:**
Stores user-generated reviews, including:
 - reviewID (Primary Key)
 - userID (Foreign Key referencing USER)
 - yelp_rest_id (Restaurant identifier)
 - rating
 - comment

Primary and foreign keys are used to maintain data integrity and enforce relationships between tables.

Additionally, the database design follows normalization principles to reduce redundancy and ensure consistency across stored data.

Restaurant-related information such as ratings, menus, and opening hours is not stored locally. Instead, it is dynamically retrieved from the Yelp API using the restaurant ID.

